

Lecture 19

Why integrable systems do not thermalise

Objectives. Course map: P5, and the first half of G4. [Lecture 0](#) began with two experiments that refused to come to equilibrium and promised that the reason was integrability. An isolated quantum system evolves unitarily, so nothing relaxes in the strict sense; what can relax is a local observable, and whether it relaxes to a thermal value depends on what the conserved charges remember. A generic chain is expected to remember only its energy and its symmetry charges; an integrable one remembers much more, and where it relaxes, the expected answer is the maximum-entropy state compatible with a complete family of its conserved charges, the generalised Gibbs ensemble. For the magnetisation after a field quench of [Lecture 18](#)'s Ising chain this is a theorem.

19.1 The question

[Lecture 0](#)'s cradle did not thermalise, and [Lecture 18](#) solved a chain with a conserved occupation for every mode. So the question is

if an isolated quantum system evolves unitarily, what relaxes, and to what?

The answer: local observables, when they relax at all, to the average in the ensemble fixed by a complete family of conserved charges ([Definition 19.5](#)); for the magnetisation after a field quench of the Ising chain this is proved ([Theorem 19.7](#)), and the Gibbs ensemble gets it wrong ([Figure 19.2](#)). The worked example is that quench.

19.2 Local and extensive charges

Densities

From here on the system is a ring of N sites, each carrying a spin $\frac{1}{2}$: the Hilbert space is $(\mathbb{C}^2)^{\otimes N}$, site $N+1$ is site 1, and τ is the translation that moves every site one step, $n \mapsto n+1$. An operator acts on a set of sites if it is the identity on every other factor.

Definition 19.1 (Local charge). A density of range r is an operator q acting on sites $1, \dots, r$. On a ring of $N \geq r$ sites its local charge is the sum of its translates,

$$Q = \sum_{n=1}^N \tau^{n-1}(q),$$

$\tau^{n-1}(q)$ acting on sites $n, \dots, n+r-1$. The range r and the density q are fixed once and

for all, and N varies: a local charge is really one operator for each chain length. A local charge that commutes with H is a *local conserved charge*.

The fixed range is the whole content: with $r = N$ allowed, any translation-invariant operator A would qualify, as the sum of the translates of A/N . The total spin $S^z = \sum_n S_n^z$ is a local charge of range 1, and so is a nearest-neighbour Hamiltonian such as [Lecture 18](#)'s Ising chain, of range 2. Local charges are *extensive*: Q is a sum of N terms of norm $\|q\|$, so $\|Q\| \leq N \|q\|$, and in a translation-invariant state $\langle Q \rangle = N \langle q \rangle$ — proportional to the length, as energy and particle number are. And local charges are rare.

Proposition 19.2 (Local charges are few). *For each fixed r , the local charges of range at most r on a ring of $N \geq r$ sites form a vector space of dimension at most 4^r , whatever N . So among the 2^N linearly independent conserved operators of a Hamiltonian with simple spectrum, the local conserved charges of range at most r span at most 4^r dimensions.*

Proof. A density of range r is an operator on $(\mathbb{C}^2)^{\otimes r}$, a space of dimension 4^r , and $q \mapsto Q$ is linear. A density of smaller range is one of range r that acts trivially on the last sites. ■

Two chains

The chain is the XXZ chain with a next-nearest-neighbour coupling,

$$H = \sum_{n=1}^N (S_n^x S_{n+1}^x + S_n^y S_{n+1}^y + \Delta S_n^z S_{n+1}^z) + J_2 \sum_{n=1}^N (S_n^x S_{n+2}^x + S_n^y S_{n+2}^y + \Delta S_n^z S_{n+2}^z). \quad (19.1)$$

At $J_2 = 0$ it is the XXZ chain, which the algebraic Bethe ansatz solves ([Lecture 15](#)'s reading); for $J_2 \neq 0$ it is not known to be integrable. [Figure 19.1](#) compares $J_2 = 0$ with $J_2 = 0.5$. The anisotropy is $\Delta = 0.55$, away from the values $\cos(\pi p/q)$ with small q at which XXZ acquires extra degeneracies.

19.3 Why integrable systems do not thermalise

Prepare a system in a pure state $|\psi_0\rangle$ that is not an eigenstate of H and let it evolve. Nothing about $|\psi(t)\rangle$ relaxes: the state stays pure and, on a finite system, almost periodic. The question is about a *local* observable O ([Definition 19.1](#)), in the thermodynamic limit taken first. Expanding in eigenstates $H|n\rangle = E_n|n\rangle$,

$$\langle \psi(t) | O | \psi(t) \rangle = \sum_n |c_n|^2 \langle n | O | n \rangle + \sum_{m \neq n} c_m^* c_n e^{-i(E_n - E_m)t} \langle m | O | n \rangle, \quad c_n = \langle n | \psi_0 \rangle. \quad (19.2)$$

Proposition 19.3 (Dephasing). *On a finite system the time average of $\langle \psi(t) | O | \psi(t) \rangle$ over $[0, T]$ converges as $T \rightarrow \infty$ to $\text{tr}(\rho_d O)$, where $\rho_d = \sum_E P_E |\psi_0\rangle \langle \psi_0| P_E$ is the initial state dephased in energy and P_E projects on the eigenspace of E . If the spectrum is non-degenerate this is the diagonal ensemble $\sum_n |c_n|^2 |n\rangle \langle n|$.*

Proof (★). The time average of $e^{-i(E_n - E_m)t}$ over $[0, T]$ is bounded in modulus by $2/(|E_n - E_m|T)$, so every term with $E_n \neq E_m$ tends to 0; the terms with $E_n = E_m$ are exactly the ones the projectors keep, and the sum is finite. ■

So whatever a local observable settles down to is fixed by ρ_d — exponentially many numbers, and no ensemble worth the name. (Not by the $|c_n|^2$ alone: inside a degenerate eigenspace the relative phases survive dephasing, and a spectator spin prepared in $|\pm x\rangle$ makes that visible.) Every relaxation statement is the claim that far fewer of those numbers matter.

The hypothesis that makes a generic system thermal

Definition 19.4 (Eigenstate thermalisation · cited: Deutsch 1991, Srednicki 1994). A Hamiltonian satisfies the *eigenstate thermalisation hypothesis* (ETH) for a local observable O if, within each sector of fixed symmetry charges (such as S^z and momentum) and in the bulk of its spectrum,

$$P_E O P_E = \left[O_{\text{mc}}(E/N) + \text{a fluctuation vanishing as } N \rightarrow \infty \right] P_E, \quad (19.3)$$

with P_E the projector on energy E within the sector and O_{mc} a smooth function — the microcanonical average in that sector at that energy density. On a non-degenerate spectrum this is the familiar $\langle n | O | n \rangle = O_{\text{mc}}(E_n/N) + \dots$; where energies are degenerate it has to be asked of the whole block, or a spectator degree of freedom defeats it with every diagonal element right.

If (19.3) holds and $|\psi_0\rangle$ lies in one symmetry sector with subextensive energy fluctuations, Proposition 19.3 averages a smooth function over a narrow window and the answer is the microcanonical one: the system *thermalises*, in the only sense a closed quantum system allows. Nothing has been proved — ETH is a hypothesis about a Hamiltonian, checked numerically model by model, and (19.3), sector by sector, is exactly the statement a figure can test (Figure 19.1).

The charges are the memory

The left panel is not an accident of the observable. A chain with about N local conserved charges, all commuting, can have its eigenstates labelled by all of them at once. What is expected to replace (19.3) there is a *generalised* eigenstate thermalisation: $\langle n | O | n \rangle$ a smooth function of the densities of a *complete* family of conserved charges — the strictly local ones for the Ising chain, quasi-local ones as well for the XXZ chain (Lecture 20's reading) · cited: Caux and Essler 2013. That reduces to a function of the energy alone only if the other densities are fixed by it, and they are not: two eigenstates at the same energy density generically carry different values of the other charges. The charges remember which state the system was prepared in, and a local observable can see the memory. The example this lecture proves is the Ising chain of Lecture 18, whose local charges are those of Proposition 19.8.

Free is linear

A quadratic Hamiltonian makes the Heisenberg equations linear: $d\gamma_k/dt = i[H, \gamma_k] = -i\varepsilon(k)\gamma_k$, so each mode turns at its own frequency and ignores the others — the quantum form of the linear flow $\dot{\varphi} = \nu$ on a torus, and the reason there is no scattering. (Lecture 18's caveats say why this is special: the Heisenberg chain's fermions interact.)

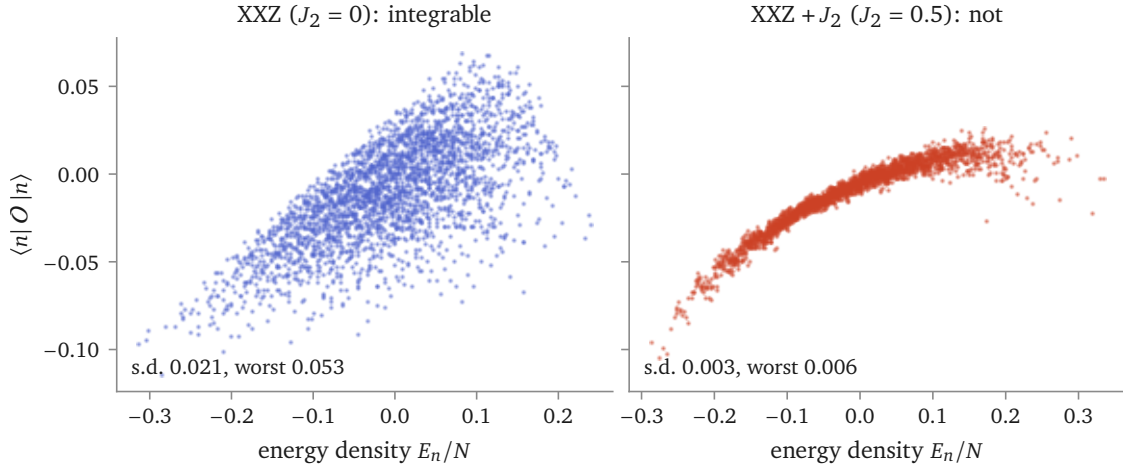


Figure 19.1: Eigenstate expectation values of the two-site operator $O = S_1^z S_2^z$ — equal, on the momentum eigenstates drawn, to the density $\frac{1}{N} \sum_n S_n^z S_{n+1}^z$, which is what is computed — in every state of one symmetry sector ($N = 18$, $S^z = 1$, $k = \pi/3$) of (19.1) at $\Delta = 0.55$: the two chains above, seen through an observable. Left, integrable, a cloud; right, with $J_2 = 0.5$, a curve. *The two numbers say different things.* Both are averages over windows of 32 consecutive states. The *standard deviation* in a window measures a typical eigenstate, and it falls in *both* models — for XXZ like $N^{-1/2}$ ($\text{sd}\sqrt{N} = 0.091, 0.091, 0.090$ at $N = 14, 16, 18$). The *largest* deviation from a window’s median measures its worst eigenstate, and there they part: its average is 0.059, 0.060, 0.053 for XXZ, roughly flat, against 0.019, 0.014, 0.006 for $J_2 = 0.5$. Three sizes cannot settle a limit; they show the two chains differing as weak eigenstate thermalisation with persistent rare states would — and a quench does not sample eigenstates typically. The sector is fixed because in a degenerate subspace $\langle n|O|n\rangle$ depends on the basis the diagonalisation happens to choose; in this sector neither the spin flip nor the reflection acts.

19.4 The generalised Gibbs ensemble, and a quench done in full

Definition 19.5 (Generalised Gibbs ensemble · cited: Rigol, Dunjko, Yurovsky and Olshanii 2007). Given conserved charges Q_a and their initial expectation values $\langle \psi_0 | Q_a | \psi_0 \rangle$, the *generalised Gibbs ensemble* is the maximum-entropy state reproducing all of them,

$$\rho_{\text{GGE}} = \frac{1}{Z} \exp \left(- \sum_a \beta_a Q_a \right), \quad (19.4)$$

the multipliers β_a being fixed by those expectation values.

With one charge, $Q_1 = H$, this is the Gibbs state and $\beta_1 = 1/T$; the question is what the others do. The claim to be tested — that a local observable relaxes to its ρ_{GGE} average — can be settled by calculation on Lecture 18’s chain.

The quench

Take the transverse-field Ising chain (18.1) at coupling J , start in the ground state at field h_0 , and at $t = 0$ change the field to h . Both Hamiltonians are quadratic in the *same* fermions

c_k : only the Bogoliubov angle moves. Write θ_k for the post-quench angle of (18.15), θ_k^0 for the pre-quench one, and

$$\chi_k = \theta_k - \theta_k^0, \quad \cos 2\chi_k = \frac{a_k a_k^0 + b_k^2}{\varepsilon_k \varepsilon_k^0}, \quad \sin 2\chi_k = \frac{4J(h_0 - h) \sin k}{\varepsilon_k \varepsilon_k^0}, \quad (19.5)$$

with $a_k = 2(h - J \cos k)$, $b_k = 2J \sin k$, $\varepsilon_k = \sqrt{a_k^2 + b_k^2}$ as in § 18.5, and a_k^0, ε_k^0 the same at h_0 . The rotation taking pre-quench modes to post-quench ones is the rotation by the *difference* angle, $\gamma_k = \cos \chi_k \gamma_k^0 - i \sin \chi_k (\gamma_{-k}^0)^\dagger$, and the pre-quench ground state is the γ^0 vacuum.

Proposition 19.6 (The occupations are the conserved data). *After the quench the post-quench mode occupations are*

$$n_k = \langle \psi(t) | \gamma_k^\dagger \gamma_k | \psi(t) \rangle = \sin^2 \chi_k, \quad (19.6)$$

independent of t , and even in k .

Proof. Each $\gamma_k^\dagger \gamma_k$ commutes with $H(h)$, so its expectation is constant; its value in the γ^0 vacuum is $|\sin \chi_k|^2$ by the rotation above. Evenness: $\cos 2\chi_k$ in (19.5) depends on k only through $\cos k$. ■

The difference angle is not a bookkeeping convention: the sum angle gives a well-formed, plausible and wrong n_k , off by as much as 0.48 for $J = 1$, $h_0 = 0.4 \rightarrow h = 2.1$.

What a local observable does

The transverse magnetisation $m^z = \frac{1}{N} \sum_n \sigma_n^z$ is the density of a one-site operator, and in a translation-invariant state it is $\langle \sigma_1^z \rangle$ exactly; (18.4) makes it $1 - \frac{2}{N} \sum_k c_k^\dagger c_k$.

Theorem 19.7 (Relaxation to the generalised Gibbs ensemble). *Let $J, h, h_0 > 0$ with $h \neq J$ and $h_0 \neq J$. For the quench above on a ring of N sites, with the NS momenta of the even sector,*

$$m^z(t) = \frac{1}{N} \sum_k \left[\cos 2\theta_k \cos 2\chi_k + \sin 2\theta_k \sin 2\chi_k \cos(2\varepsilon_k t) \right]. \quad (19.7)$$

In the thermodynamic limit the sum becomes $\int dk/2\pi$, the oscillating term tends to 0 as $t \rightarrow \infty$ like $t^{-3/2}$, and the limit is

$$m^z(\infty) = \int \frac{dk}{2\pi} \cos 2\theta_k (1 - 2n_k) = \text{tr}(\rho_{\text{GGE}} m^z), \quad (19.8)$$

the average in the ensemble (19.4) built from the occupations (19.6).

Proof. The ground state at $h_0 > 0$ is in the even sector — in the σ^x basis the field gives non-positive off-diagonal elements connecting every configuration, so Perron–Frobenius makes the ground vector unique and strictly positive, and Π only permutes those configurations — so the NS grid is the right one. Each pair $(k, -k)$ then spans a two-dimensional even-parity space, and the expectation of $c_k^\dagger c_k$ in it is $\frac{1}{2}(1 - \cos 2\theta_k \cos 2\chi_k) - \frac{1}{2} \sin 2\theta_k \sin 2\chi_k \cos 2\varepsilon_k t$; summing gives (19.7). Every term is even in k , since $\cos 2\theta$ and $\cos 2\chi$ are even and $\sin 2\theta$ and $\sin 2\chi$ are odd.

For the limit, $\varepsilon'(k) \propto \sin k$ vanishes exactly at $k = 0$ and $k = \pi$ and nowhere else, and the amplitude is

$$\sin 2\theta_k \sin 2\chi_k = \frac{8J^2(h_0 - h) \sin^2 k}{\varepsilon_k^2 \varepsilon_k^0}. \quad (19.9)$$

This is where the hypotheses are spent. With $h \neq J$ and $h_0 \neq J$ neither ε vanishes anywhere, so (19.9) has a double zero at both stationary points, and stationary phase with an amplitude vanishing to second order gives $t^{-3/2}$ — not the generic $t^{-1/2}$ of a non-vanishing amplitude. What is left is $\int \cos 2\theta_k \cos 2\chi_k$, and $\cos 2\chi_k = 1 - 2n_k$ by (19.6). The ensemble (19.4) built from the charges $Q[f]$ below is diagonal in the occupations, so its modes are independent with occupations n_k and the anomalous correlator $\langle \gamma_k^\dagger \gamma_{-k}^\dagger \rangle$ vanishes; the two numbers agree. ■

The hypotheses are spent there: at $h_0 = J$ the zero of (19.9) is simple and the decay is t^{-1} , at $h = 0$ the dispersion is flat and nothing decays, and at $h_0 = 0$ there is no single ground state.

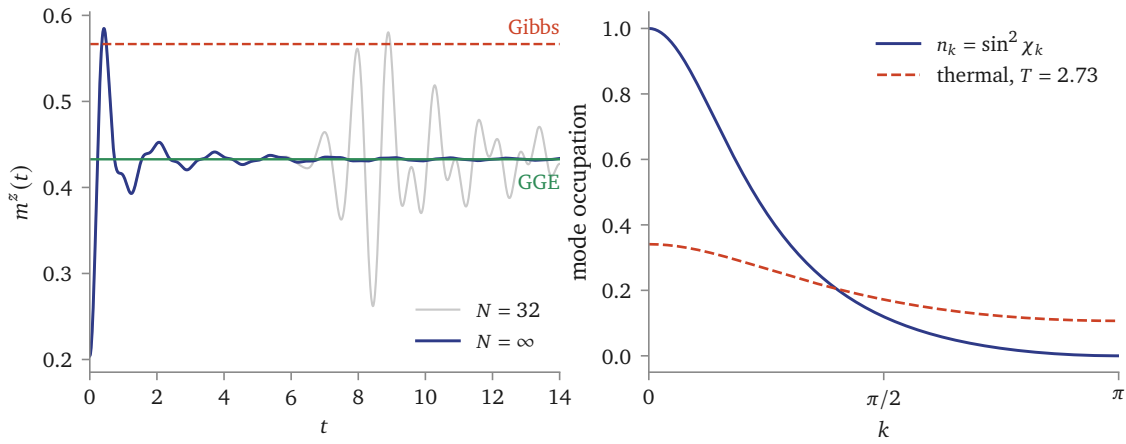


Figure 19.2: The quench $J = 1$, $h_0 = 0.4 \rightarrow h = 1.9$. *Left*: $m^z(t)$ from (19.7). In the thermodynamic limit (dark) it settles onto the generalised Gibbs value 0.4327; the Gibbs value at the temperature $T = 2.730$ carrying the same energy is 0.5667. At $N = 32$ (pale) nothing decays — the sum revives, and only its time average converges. *Right*: the conserved $n_k = \sin^2 \chi_k$ against the thermal occupations of the same energy; the closest Fermi–Dirac fit, at $T = 3.24$, is out by 0.243 root mean square. The left panel’s formula is checked against exact diagonalisation of the spin chain at $N = 8, 10$, the initial state’s parity included.

Theorem 19.7 takes $N \rightarrow \infty$ first and $t \rightarrow \infty$ second, and the two do not commute: at finite N the sum (19.7) revives, as the pale curve of Figure 19.2 does, and only its time average converges.

Which combinations of the n_k are local

Definition 19.5 asks for charges, and a single n_k is not an observable of the chain: it is a sum over all N sites whose coefficients never decay. Lecture 18’s reading built the local charges by weighting the k -block of (18.13); this is that weighting.

Proposition 19.8 (The local charges of the Ising chain · sketched). Let $J, h > 0$ with $h \neq J$, and let f be an even function on the Brillouin zone. Write $Q[f] = \sum_k f(k)(n_k - \frac{1}{2})$. Then $Q[f]$ is a sum of fermion bilinears whose real-space coefficients are the Fourier modes of $f \cos 2\theta$ and $f \sin 2\theta$. Hence

$$Q[f] \text{ has finite range} \iff f = \varepsilon \cdot (\text{a trigonometric polynomial}), \quad (19.10)$$

and $f = \varepsilon \cos mk$ gives a charge of range $m + 2$ in the spins.

Proof sketch (★). In the pair basis $Q[f] = \sum_{0 < k < \pi} \Psi_k^\dagger (f h_k / \varepsilon_k) \Psi_k$, so the hopping and pairing functions are $A = f a_k / \varepsilon_k$ and $B = f b_k / \varepsilon_k$. Finite range means both are trigonometric polynomials, and a_k, b_k are. For the converse, $g = f / \varepsilon$ is recovered from A and B by one identity,

$$(h + J \cos k) a_k - J \sin k b_k = 2(h^2 - J^2), \quad \text{so} \quad g = \frac{(h + J \cos k)A - J \sin k B}{2(h^2 - J^2)}, \quad (19.11)$$

a trigonometric polynomial whenever A and B are — and this is exactly where $h \neq J$ is spent. Taking $f = \varepsilon$ returns H itself; $f = \varepsilon \cos k$ returns $-\frac{1}{2}Q_+$ of (18.24) up to a constant, checked against the even-sector ground state at $N = 8, 10$, where the sign is visible. ■

The ε weight is not decoration. Drop it, and $Q[\cos mk]$ is not local: its coefficients decay exponentially, at the rate $|\ln(h/J)|$ — the branch points of $1/\varepsilon$ sit at $z = e^{ik} = h/J$ and $z = J/h$, and whichever is nearer the unit circle sets the rate — without ever vanishing. Such a charge is *quasi-local*, the class that Lecture 20's reading shows the XXZ chain needs. Both ends of the range are special: at $h = J$ the two branch points meet on the circle and the coefficients decay only like $1/d$, while at $h = 0$, where $\varepsilon = 2J$ is constant, the weight is a number and every trigonometric polynomial is strictly local again.

None of this weakens Theorem 19.7: knowing every $Q[\varepsilon P]$ is knowing every Fourier coefficient of $\varepsilon(k)n(k)$, and $\varepsilon > 0$ almost everywhere, so the strictly local charges fix n_k almost everywhere and with it every local observable's relaxed value. Numerically, keeping $Q[\varepsilon \cos mk]$ for $m \leq M$ and maximising the entropy gives m^z to 1.7×10^{-3} at $M = 4$ and 8×10^{-8} at $M = 16$ — and at $M = 0$, where only the energy is kept, that is the Gibbs ensemble, with exactly the error of Figure 19.2.

19.5 What it buys

- *Lecture 0's question, answered in one model*: after a quench the Ising chain's magnetisation relaxes, and to the generalised Gibbs value, not the Gibbs one (Theorem 19.7, Figure 19.2).
- *Why*: the occupations n_k are conserved (Proposition 19.6), and the local charges fix them (Proposition 19.8); a local observable remembers them.
- *The ensemble* that Lecture 20 carries to the interacting gas, where one function, the rapidity density, plays the part of the n_k .

19.6 Caveats, in one place

- *Status*. Proposition 19.3 (proof in a reading), Proposition 19.6 and Theorem 19.7 are proved; Proposition 19.8 is sketched. ETH is a hypothesis, checked numerically; generalised ETH is cited; the GGE's definition is cited.

- *The hypotheses of Theorem 19.7 are spent.* At $h_0 = J$ the decay is slower, at $h = 0$ it does not happen, at $h_0 = 0$ there is no single ground state, and at $h = J$ the observable m^z cannot tell the Gibbs and generalised ensembles apart.
- *The order of the limits:* $N \rightarrow \infty$ first, then $t \rightarrow \infty$. At finite N the magnetisation revives, and only its time average converges.
- *Which charges.* For the Ising chain the strictly local charges fix the occupations almost everywhere (Proposition 19.8); for the XXZ chain the ultralocal charges are not enough, and quasi-local ones are needed (Lecture 20's reading).
- *Every statement about a non-integrable chain is numerical,* and a state that carries a thermal state's charge densities relaxes to thermal values even in an integrable chain.

Reading

19.7 This was known in 1970

Remark 19.9 (This was known in 1970). Barouch, McCoy and Dresden computed the magnetisation of the XY chain after a field quench in 1970 and found it non-ergodic: it does not approach the equilibrium value. The ensemble that explains it was written down thirty-seven years later.

Summary.

- *Local charges* (Definition 19.1): fixed range, extensive, and few (Proposition 19.2).
- *Dephasing* (Proposition 19.3): a local observable's time average is fixed by the energy-dephased state (the diagonal ensemble when the spectrum is non-degenerate); ETH (Definition 19.4), sector by sector, makes it thermal in a generic chain.
- *The GGE* (Definition 19.5): maximum entropy given all the charges.
- *The Ising quench* (Proposition 19.6 and Theorem 19.7): the occupations are conserved, the magnetisation relaxes like $t^{-3/2}$ to the GGE value, and the Gibbs value at the same energy is different.
- *Next:* Lecture 20 lets the ensemble vary slowly in space, and explains the cradle.

Tutorial. Exercise Sheet 19.

References

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